

JavaScript Advanced Topics 1. Core Advanced Concepts - Closures - Scope (Global, Local, Block scope) - Hoisting - Execution Context & Call Stack - Event Loop (Microtask & Macrotask queue) - This keyword behavior 2. Functions & Advanced Patterns - First-class functions - Higher-order functions - Callback functions - Function currying - Function composition - Recursion - IIFE 3. Asynchronous JavaScript - Callbacks - Promises - async / await - Error handling - Fetch API / AJAX - Parallel vs Sequential execution 4. Objects & OOP - Object creation patterns - Prototypes & Prototype chain - Classes (ES6) - Inheritance - Encapsulation - Polymorphism 5. ES6+ Features - let / const vs var - Arrow functions - Template literals - Destructuring - Spread & Rest - Modules - Optional chaining - Nullish coalescing 6. Advanced Data Handling - Map, Set, WeakMap, WeakSet - Deep vs Shallow copy - JSON handling - Immutable data 7. DOM & Browser APIs - Event delegation - Debouncing & Throttling - Intersection Observer - Mutation Observer - LocalStorage / SessionStorage - Cookies 8. Error Handling & Debugging - try/catch/finally - Custom errors - Debugging - Performance optimization 9. Functional Programming - Pure functions - Immutability - Side effects - map, filter, reduce 10. Interview Level Concepts - Memory management - Event bubbling & capturing - Deep vs shallow equality - Currying vs partial application - Polyfills 11. Real Applications - Modular architecture - Code splitting - Lazy loading - Design patterns 12. Modern Ecosystem - Webpack / Vite - Babel - TypeScript - Testing (Jest)